Xiangyi Gao

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Research interests: computer vision, robot perception, and learning systems that act in the physical world.

EDUCATION

University of North Carolina at Chapel Hill — Chapel Hill, NC
B.S. in Computer Science | *GPA: 4.0/4.0* — *Expected May 2029*

Relevant coursework: Data Structures, Discrete Mathematics, Differential Equations, Multivariable Calculus; currently studying Systems, Foundations of Programming, Models of Computation, and Linear Algebra.

RESEARCH EXPERIENCE

Bai Lab, UNC-Chapel Hill — Chapel Hill, NC
Undergraduate Research Assistant — *Jan. 2026 – May 2026*

- Varied ionic-liquid and monomer ratios in ionogel formulations; analyzed conductivity and mechanical stress–strain measurements.
- Investigated copper-electrode oxidation drift, helped transition to carbon electrodes, and collected EMG/EEG and motion-response data for a wearable-patch manuscript submitted to *Advanced Materials Technologies*.

State Key Laboratory of Crystal Materials, Shandong University — Jinan, China
Research Assistant — *Dec. 2022 – Oct. 2023*

- Synthesized rare-earth-substituted $Bi_5Ti_3FeO_{15}$ ceramics and characterized phase and microstructure with XRD and SEM; second author on two peer-reviewed papers (below).

PUBLICATIONS

- X.-Y. Yu, **X.-Y. Gao**, Q. Wang, H.-L. Li, Y.-J. Wan, E.-M. Liang, C.-M. Wang. "Lutetium-substituted high-T_C $Bi_5Ti_3FeO_{15}$ exhibiting superior piezoelectric performance." *Ceramics International*, 51, 2025.
- X.-Y. Yu, **X.-Y. Gao**, Q. Wang, H.-L. Li, Y.-J. Wan, E.-M. Liang, C.-M. Wang. "Superior piezoelectric performance in ytterbium-substituted high-T_C $Bi_5Ti_3FeO_{15}$." *Journal of Alloys and Compounds*, 1010, 2025.

SELECTED PROJECTS

SportLens: Transfer Learning for Sports Image Classification — Sept. 2026
Independent computer vision project | *PyTorch, ResNet-18*

- Compared head-only training with partial fine-tuning (layer4 + head) of ResNet-18 on 100 sports classes; reached 94.0% test accuracy vs. 91.8% for a matched head-only control, and documented failure cases on external images.

DigitShift: Translation Robustness on MNIST — Sept. 2026
Independent machine learning project | *scikit-learn*

- Showed a two-pixel translation drops a random-forest classifier from 96.95% to 73.01–77.73% on the 10,000-image MNIST test set; four-direction augmentation restored 96.20–97.24%.

MineJob: AI Job-Application Workshop (HackNC 2026) — Oct. 2026
Team hackathon project | *Three.js, JavaScript, FastAPI*

- Built the explorable Three.js 3D front end (voxel scene, character, camera controls, accessible HTML fallback) and integrated it with the FastAPI resume and cover-letter generation backend.

Lumina: Automata Theory Workbench — Oct. 2026
Independent web app | *TypeScript, Next.js, React Flow*

- DFA/NFA/PDA/CFG workbench with animated runs, subset construction, minimization, CFG-to-PDA, CNF, and a practice mode; 133 tests; live demo.

LEADERSHIP, PATENTS & ADDITIONAL EXPERIENCE

FIRST Tech Challenge Team 23755 — Jinan, China
Founder and Lead Programmer — *Sept. 2023 – Dec. 2024*

- Led a 10-member team to an Inspire Award; programmed autonomous navigation and PID control in Java.

Patents: Sole inventor, two granted Chinese patents on a precision pill dispenser: CN 119503293 B (invention), CN 223534063 U (utility model).

TECHNICAL SKILLS & HONORS

Programming and tools: Python (PyTorch, NumPy, scikit-learn), C, Java, JavaScript, TypeScript, Git, Linux, LaTeX.
Honors: Two-time AIME qualifier; BPhO Gold; HiMCM Honorable Mention; WorldQuant Gold; Stanford ML Specialization.